

Chapter 1

Exercise 1.

Condition to maintain arbitrage:

$$0 < d < 1 + r < u$$

If $X_0 = 0$,

$$X_1 = \Delta_0 S_1 + (1 + r)(X_0 - \Delta_0 S_0) \implies X_1 = \Delta_0 S_1 + (1 + r)(-\Delta_0 S_0)$$

$$X_1(H) = \Delta_0 u S_0 + (1 + r)(-\Delta_0 S_0) \implies X_1(H) = \Delta_0 S_0(u - 1 - r)$$

$$X_1(T) = \Delta_0 d S_0 + (1 + r)(-\Delta_0 S_0) \implies X_1(T) = \Delta_0 S_0(d - 1 - r)$$

As per the condition above:

$$(u - (1 + r)) > 0$$

$$d - (1 + r) < 0$$

As a result the wealth at $T=1$ will be of opposite signs:

$$X_1(H)X_1(T) < 0$$

Exercise 2.

At time $T=1$:

$$X_1 = \Delta_0 S_1 + \Gamma_0(S_1 - 5)^+ - \frac{5}{4}(4\Delta_0 + 1.20\Gamma_0)$$

$$S(0) = 4, u = 2, d = \frac{1}{2}, r = \frac{1}{4}, \Delta_0 = \frac{1}{2}$$

$$X_1(H) = \Delta_0 8 + \Gamma_0(8 - 5)^+ - \frac{5}{4}(4\Delta_0 + 1.20\Gamma_0) \implies X_1(H) = 3\Delta_0 + 1.5\Gamma_0$$

$$X_1(T) = \Delta_0 2 + \Gamma_0(2 - 5)^+ - \frac{5}{4}(4\Delta_0 + 1.20\Gamma_0) \implies X_1(T) = -(3\Delta_0 + 1.5\Gamma_0)$$

$X_1(H)$ and $X_1(T)$ are same magnitude and opposite signs. So there is positive probability that wealth is positive and positive probability that wealth is negative at $T=1$

Exercise 3.

$$V_0 = \frac{1}{1 + r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)]$$

$$V_0 = \frac{1}{1+r}[\tilde{p}uS_0 + (1-\tilde{p})dS_0] \implies \frac{S_0}{1+r}[\tilde{p}(u-d) + d]$$

$$\tilde{p} = (1+r-d)/(u-d)$$

$$V_0 = S_0$$

Exercise 4

We need to show (1):

$$X_{n+1}(\omega_1\omega_2\omega_3 \dots \omega_n T) = V_{n+1}(\omega_1\omega_2\omega_3 \dots \omega_n T)$$

We take initial wealth as V_0

$$X_0 = V_0$$

We assume the above equation 1 is valid for some $n \leq N$,
we need to show it is valid for $n+1$:

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n)$$

$$X_{n+1}(T) = \Delta_n S_{n+1}(T) + (1+r)(X_n - \Delta_n S_n) \implies \Delta_n dS_n + (1+r)(X_n - \Delta_n S_n)$$

$$\implies d\Delta_n S_n + (1+r)(X_n - \Delta_n S_n) = \Delta_n S_n(d-1-r) + X_n(1+r)$$

$$\Delta_n = \frac{V_{n+1}(H) - V_{n+1}(T)}{(u-d)S_n}$$

$$X_{n+1}(T) = \frac{V_{n+1}(H) - V_{n+1}(T)}{u-d}(d-1-r) + V_n(1+r) \implies (V_{n+1}(T) - V_{n+1}(H))\tilde{p} + V_n(1+r)$$

$$V_n = \frac{1}{1+r}[\tilde{p}V_{n+1}(H) - V_{n+1}(T)\tilde{q}]$$

$$X_{n+1}(T) = [V_{n+1}(T) - V_{n+1}(H)]\tilde{p} + [\tilde{p}V_{n+1}(H) + V_{n+1}(T)\tilde{q}] = V_{n+1}(T)$$

Exercise 5.

$$S(0) = 4, u = 2, d = \frac{1}{2}, r = \frac{1}{4}, \Delta_0 = 0.1733$$

At $T=1$,

$$\Delta_1 = \frac{V_2(H) - V_2(T)}{(u-d)S_1} = (3.20 - 2.4)/(1.5 * 8) = 0.067$$

Using the wealth equation for $T=2$:

$$X_2 = \Delta_1 S_2 + (1+r)(X_1 - \Delta_1 S_1)$$

$$X_2(H) = \Delta_1 u S_1 + (1+r)(X_1 - \Delta_1 S_1) = 0.067 * 16 + (5/4)(2.24 - 0.067 * 8) = 1.07 + 2.13 = 3.20$$

$$X_2(T) = \Delta_1 d S_1 + (1+r)(X_1 - \Delta_1 S_1) = 0.067 * 4 + 1.25 * (2.24 - 0.067 * 8) = 0.27 + 2.13 = 2.40$$

Similar calculations can be done for $T=2$

Exercise 6.

$$S(0) = 4, u = 2, d = \frac{1}{2}, r = \frac{1}{4}, K = 5, X_1 = 1.5$$

$$X_1(H) = \Delta'_0 u S_0 + (1+r)(0 - \Delta'_0 S_0) + C$$

$$X_1(T) = \Delta'_0 d S_0 + (1+r)(0 - \Delta'_0 S_0) + C$$

Bank can hedge away the long call position:

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{(u - d)S_0} = 3/6 = 0.5$$

$$C = 0.5S_1 + (1+r)(1.20 - 0.5S_0)$$

$$X_1(H) = \Delta'_0 u S_0 + (1+r)(0 - \Delta'_0 S_0) + 0.5u S_0 + (1+r)(1.20 - 0.5S_0) \Rightarrow u S_0(\Delta'_0 + 0.5) + (1+r)(1.20 - S_0(\Delta'_0 + 0.5))$$

$$X_1(T) = \Delta'_0 d S_0 + (1+r)(0 - \Delta'_0 S_0) + 0.5d S_0 + (1+r)(1.20 - 0.5S_0) \Rightarrow d S_0(\Delta'_0 + 0.5) + (1+r)(1.20 - S_0(\Delta'_0 + 0.5))$$

The only solution to these equations is

$$\Delta'_0 = -0.5$$

We will short sell 0.5 units of S at T=0 and get 2 in cash. We will then invest this in money market account to yield $1.25 \times 2 = 2.5$ at T=1.

In both H and T scenarios we will get 1.5 if we include the pay off from the Call Option.

Exercise 7.

Similar to previous question. The trader should short -0.1733 shares at time T=0. Cash received should be reinvested in money market account for T=1:

$$0.1733 \times 4 = 0.693$$

At T=1,

If H then Stock is at 8. Option Value is 2.24, Total pay off is:

$$0.693 \times 1.25 + 2.24 - 0.1733 \times 8 = 1.72$$

If T, then Stock is at 2. Option Value is 1.2, Total pay off is:

$$0.693 \times 1.25 + 1.2 - 0.1733 \times 2 = 1.72$$

We earned risk free rate for T=1. We will again take delta position for T=2 and so on till T=3.

Exercise 8.

$$S(0) = 4, u = 2, d = \frac{1}{2}, r = \frac{1}{4}, K = 4$$

$$v_3(s, y) = \left(\frac{y}{4} - K\right)^+$$

$$v_n(s_n, y_n) = \frac{1}{1+r} \left[\tilde{p} v_{n+1}\left(2s_n, \frac{(n * y_n + 2s_n)}{n+1}\right) + \tilde{q} v_{n+1}\left(0.5s_n, \frac{(n * y_n + 0.5s_n)}{n+1}\right) \right]$$

$$v_n(s_n, y_n) = \frac{2}{5} [v_{n+1}(2s_n, \frac{(ny_n + 2s_n)}{n+1}) + v_{n+1}(0.5s_n, \frac{(ny_n + 0.5s_n)}{n+1})]$$

$$v_2(s_2, y_2) = \frac{2}{5} [v_3(2s_2, \frac{(2y_2 + 2s_2)}{3}) + v_3(0.5s_2, \frac{(2y_2 + 0.5s_2)}{3})]$$

The number of possible values of s and y are:

HHH: 32, y = 60

HHT: 8, y = 4+8+16+8=36

HTH: 8, y = 4+8+4+8=24

HTT: 2, y = 4+8+4+2=18

TTH: 8, y = 4+2+4+8=18

THT: 2, y = 4+2+4+2=12

TTT: 2, y = 4+2+1+2=9

TTT: 0.5, y = 4+2+1+0.5=7.5

$$v_3(32,60) = 60/4 - 4 = 11$$

$$v_3(8,36) = 36/4 - 4 = 5$$

$$v_3(8,24) = 24/4 - 4 = 2$$

$$v_3(2,18) = 18/4 - 4 = 0.5$$

$$v_3(8,18) = 18/4 - 4 = 0.5$$

$$v_3(2,12) = 0$$

$$v_3(2,9) = 0$$

$$v_3(0.5,7.5) = 0$$

At T=2, possible combinations(s, y):

HH: (16, 4+8+16=28)

HT: (4, 4+ 8 + 4=16)

TH: (4, 4+2+4 =10)

TT: (1, 4+2+1 =7)

$$v_2(16,28) = 2/5[v_3(32,60) + v_3(8,36)] = 0.4 * 16 = 6.4$$

$$v_2(4,16) = 2/5[v_3(8,24) + v_3(2,18)] = 0.4 * 2.5 = 1$$

$$v_2(4,10) = 2/5[v_3(8,18) + v_3(2,12)] = 0.4 * 0.5 = 0.2$$

$$v_2(1,7) = 2/5[v_3(2,9) + v_3(0.5,7.5)] = 0$$

At T=1,

H: (8, 12)

T: (2, 6)

$$v_1(8,12) = 2/5[v_2(16,28) + v_2(4,16)] = 0.4 * [6.4+1] = 2.96$$

$$v_1(2,6) = 2/5[v_2(4,10) + v_2(1,7)] = 0.4 * [0.2] = 0.08$$

$$v_0 = 2/5[v_1(8,12) + v_1(2,6)] = 0.4 [2.96+0.08] = 1.216$$

$$\delta_n = \frac{v_{n+1}(2s_n, y_n + 2s_n) - v_{n+1}(0.5s_n, y_n + 0.5s_n)}{1.5s_n}$$

Exercise 9.

1.

$$X_{n-1} = \Delta_{n-1}S_1 + (1+r)(S_{n-1} - \Delta_{n-1}S_{n-1})$$

Here u, d and r are dependent on coin tosses

$$X_n = V_n$$

$$X_{n-1} = \Delta_{n-1}u_{n-1}S_{n-1} + (1+r_{n-1})(S_{n-1} - \Delta_{n-1}S_{n-1})$$

$$X_{n-1} = \Delta_{n-1}d_{n-1}S_{n-1} + (1+r_{n-1})(S_{n-1} - \Delta_{n-1}S_{n-1})$$

$$\Delta_{n-1} = \frac{V_n(H) - V_n(T)}{(u_{n-1} - d_{n-1})S_{n-1}} \Rightarrow \frac{X_n(H) - X_n(T)}{(u_{n-1} - d_{n-1})S_{n-1}}$$

Solving similar to the book

$$\tilde{p}_{n-1} = \frac{1 + r_{n-1} - d_{n-1}}{u_{n-1} - d_{n-1}}$$

$$2. \delta_n = \frac{V_n(\omega_1 \dots \omega_{n-1}H) - V_n(\omega_1 \dots \omega_{n-1}T)}{S_n(\omega_1 \dots \omega_{n-1}H) - S_n(\omega_1 \dots \omega_{n-1}T)}$$

$$3. S_0 = 80, u = +10, d = -10, K = 80$$

At T=4,

$$V_4 = [\tilde{p}_4 V_5(H) + (1 - \tilde{p}_4) V_5(T)]$$

$$\tilde{p}_4 = \frac{1 - (110/120)}{130/120 - 110/120} = 10/20 = 0.5$$

$$V_4 = 0.5 * 50 + 0.5 * 30 = 40$$

Similarly we can proceed as below, we get the price as 9.375



